

SDS-PAGE gels preparation, running and staining

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Materials and Equipment

Glass plate assembly

- Glass plate.
- Spacers 2x.
- Alumina plate (white plate).
- Plastic plate.
- Comb (10 lanes or 15 lanes).

Multiple gel caster assembly

- Multiple gel caster chamber.
- Red rubber piece large.
- Red rubber piece small 2x.
- Gray rubber tube piece.
- Assembled glass plates 8x.
- Back plate small.
- Back plate large.
- Spring clamps 4x.



SDS-PAGE gel components

- Acrylamide/Bis-acrylamide, 30% solution.
- 1.5 M Tris pH 8.8.
- 1 M Tris pH 6.8.
- 10% SDS solution.
- 10% APS solution.
- TEMED.
- (Optional) Isopropanol.

Electrophoresis unit setup

- Gel holder (max 2x gels).
- Buffer chamber.
- Spring clamps 2x per gel.
- Buffer chamber lid with power cables.
- Power station.
- 1x Electrophoresis Running Buffer (TGS; Tris, Glycine, SDS).
- Pro Blue Safe staining solution.
- 4x SDS-PAGE sample buffer (SB)
- PageRuler pre-stained protein ladder.
- Microwave.
- Gel-imager (gray tray).



Method

SDS-PAGE gel preparation

1. Wash all components of the multiple gel caster and all the plates before use.
2. Place the rubber pieces in the multiple gel caster to prevent leaking (red rubber piece large, small (2x) and gray rubber tube piece).
3. Place a plastic plate against the back of the multiple gel caster, to make the disassembly of the glass plates easier after gel casting.
4. Assemble 8x glass plates (glass plate – spacer 2x – alumina plate – plastic plate) and place them in the multiple gel caster with the glass plate towards the back of the multiple gel caster.
5. To tighten the assembled glass plates together, place an extra glass plate and plastic plate in the end and close the multiple caster with the back plate small and back plate large.
6. Use 4x spring clamps to secure the plates in the multiple gel caster.
7. Fill the multiple gel caster with water and wait 10 minutes to check if leaking occurs.
8. Discard the water carefully (plates can fall out of the multiple gel caster!) and prepare the separating gel in the fume hood.

SDS-PAGE gel recipes

■ Separating Gel 12%:		■ Separating Gel 15%:	
Demineralized Water	16.5 mL	Demineralized Water	11.5 mL
1.5 M Tris pH 8.8	12.5 mL	1.5 M Tris pH 8.8	12.5 mL
Acrylamide solution	20.0 mL	Acrylamide solution	25.0 mL
10% SDS	0.5 mL	10% SDS	0.5 mL
10% APS	0.5 mL	10% APS	0.5 mL
TEMED	20 µL	TEMED	20 µL
■ Stacking Gel 4%:		■ 4x SDS-PAGE sample buffer stock	
Demineralized Water	27.2 mL	1 M Tris pH 6.8	2.5 mL
1 M Tris pH 6.8	5.0 mL	SDS pellets	1.0 g
Acrylamide solution	6.8 mL	0.1% BPB	0.8 mL
10% SDS	0.4 mL	Glycerol (98%)	4.0 mL
10% APS	0.4 mL	Demineralized Water	1.7 mL
TEMED	40 µL		

9. Add the components of the separating gel, except the 10% APS and TEMED, together in a 50 mL Falcon tube and mix.
10. To solidify the gel add 10% APS and TEMED to the separating gel mixture and mix thoroughly.
11. Quickly pipet or decant the separating gel mixture between the glass plates, but leave enough space on top for the stacking gel with the comb (discard pipet tips and paper towels contaminated with acrylamide in the blue waste box!).
12. Add demineralized water on top of the separating gel to prevent oxidation of the top layer and remove any air bubbles (Isopropanol can also be used for this).
13. Wait until the separating gel is solidified (30-45 minutes), then discard the water (or isopropanol) carefully and prepare the stacking gel in the fume hood.

14. Mix the stacking gel mixture thoroughly and pipet or decant the stacking gel mixture on top of the separately gel.
15. Immediately place the combs (10 lanes or 15 lanes) in the stacking gel.
16. Wait till the stacking gel is solidified (30-45 minutes), and carefully remove each glass plate with gel from the multiple gel caster.
17. Pack each glass plate with gel in a wet paper towel and place all the gels in aluminum foil or a plastic back, mark it and put it in the fridge (4°C) for future use.

SDS-PAGE electrophoresis and staining

1. Place one or two gels in the gel holder with 2x spring clamps and place it in the buffer chamber.
2. Fill the space between the glass plate and the gel holder with 1x electrophoresis running buffer and also fill one third of the buffer chamber with 1x electrophoresis running buffer.
 - 10x electrophoresis running buffer (TGS) stock preparation:
 - 250 mM Tris-Base (121.14 g/mol) => 30.3 g/L.
 - 1.92 M Glycine (75.07 g/mol) => 144 g/L
 - 1% w/v SDS (288.38 g/mol) => 10 g/L
3. Carefully remove the comb from the gel and wash the lanes of the gel by pipetting buffer.
4. Load the samples (max 45 µL with 10 lanes and 25 µL with 15 lanes) in 1x SDS-PAGE sample buffer and the pre-stained protein ladder (≈ 4 µL) on the gel.
5. Place the lid on the electrophoresis unit and make sure the power cables are correctly connected to the unit and the power station.
6. Switch on the power station and run the gel on 120 V (or 35 mA) for 45 – 60 minutes.
7. After running the gel, disassemble the glass plate and carefully transfer the gel to a box filled with water and clean several times with water.
8. Boil the gel in water in a microwave at 600 W for 2-3 minutes to remove the SDS, then discard the water and add Pro Blue Safe staining solution and boil again at 600 W for 1-2 minutes to stain the gel.
9. To de-stain the gel remove the staining solution (transfer to falcon tube for re-use) and clean several times with water and boil the gel in water again at 600 W for 2-3 minutes.
10. Take picture of the gel with the gel-imager on the gray-tray and discard the gel in the blue waste box.